SOFTWARE REQUIREMENTS SPECIFICATION (SRS)
For
TrustyHands – A Premium Service Marketplace

Prepared by -

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Submitted to :

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a comprehensive and detailed description of the “TrustyHands” system. This document outlines the functional and non-functional requirements, system architecture, and operational constraints of the platform. It serves as a reference for developers, stakeholders, and evaluators to understand the design and behavior of the system.

1.2 Scope

TrustyHands is a web-based service marketplace designed to connect customers with verified local service providers such as electricians, plumbers, and cleaners. The platform ensures reliability through a worker verification system and enhances user experience through intelligent search, booking, and communication features.
The system supports three primary user roles:

- Customers who search and book services
- Workers who provide services and manage requests
- Admin who ensures system integrity and verifies users

1.3 Definitions and Abbreviations

- SRS – Software Requirements Specification
- JWT – JSON Web Token
- MERN – MongoDB, Express.js, React.js, Node.js
- CRUD – Create, Read, Update, Delete
- API – Application Programming Interface

2. Overall Description

2.1 Product Perspective

TrustyHands is a standalone full-stack web application developed using the MERN stack. It follows a client-server architecture where the frontend interacts with backend services through RESTful APIs. The backend processes requests and stores data in a MongoDB database.

2.2 User Classes and Characteristics

Customer:

Customers are general users who want to hire local service providers. They require a simple interface to search, book, and review workers.

Worker:

Workers are skilled professionals who register on the platform to provide services. They must create detailed profiles and undergo verification before becoming active.

Admin:

Admins are system managers responsible for maintaining platform integrity, approving workers, and monitoring activities.

2.3 Operating Environment

- Client Side: Modern browsers such as Chrome, Firefox, Edge, Safari
- Server Side: Node.js runtime environment
- Database: MongoDB (Local / Atlas)

3. System Architecture

The system follows a layered architecture:

- Presentation Layer (React.js UI)
- Application Layer (Express.js API)
- Data Layer (MongoDB)

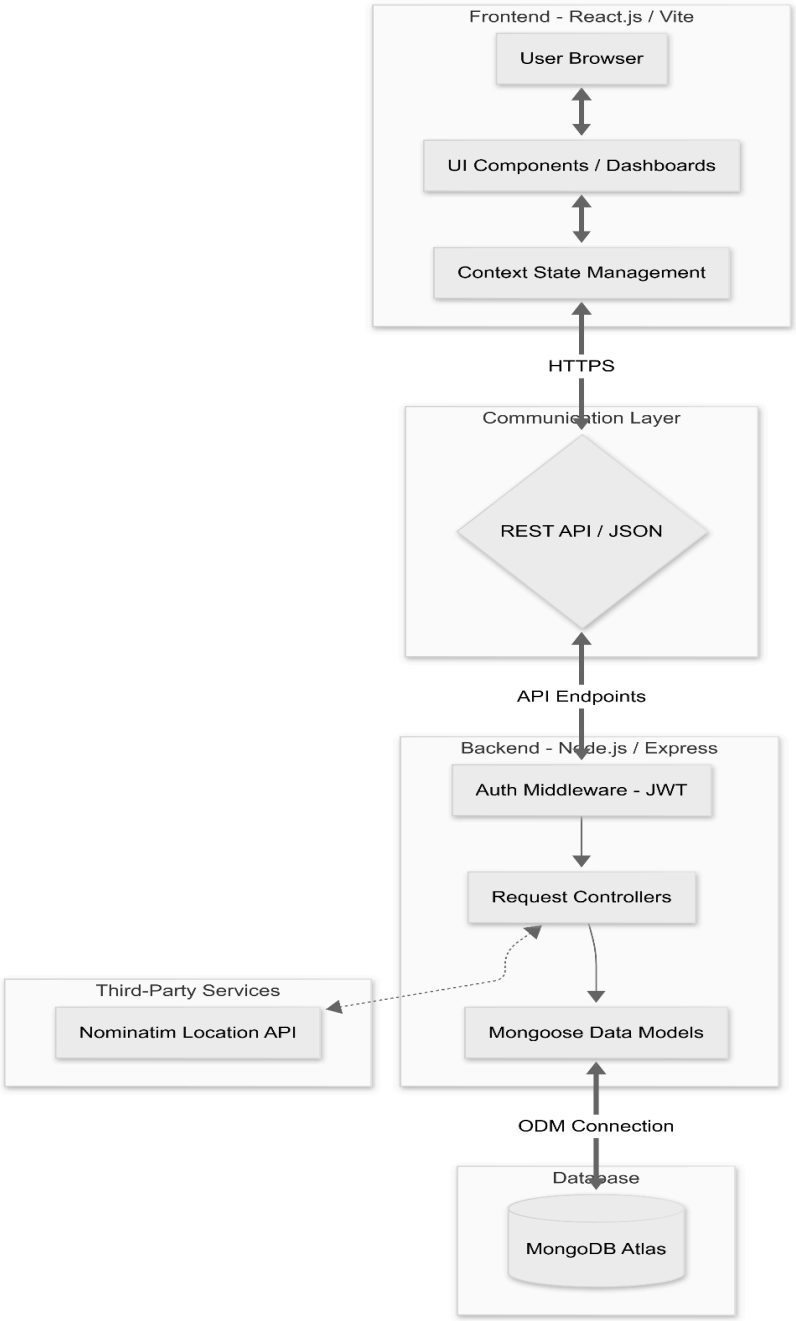


Figure 1: Overall System Architecture of TrustyHands Platform

4. System Features

4.1 User Authentication and Profile Management

The system provides secure authentication using JWT tokens. Users can register through a multi-step onboarding process tailored for customers and workers. Passwords are securely stored using hashing techniques. Users can update profiles, upload images, and manage personal details.

4.2 Service Search and Booking System

Customers can search for workers based on skills and location. The system supports dynamic filtering and displays only verified workers. Customers can send booking requests, which workers can accept or reject.

4.3 Worker Dashboard

Workers can manage their profiles, update skills, and handle job requests. They can view booking history and customer feedback. Worker accounts remain inactive until approved by the admin.

4.4 Admin Panel

The admin panel provides control over the system. Admins can approve or reject worker applications, manage users, and monitor system performance.

4.5 Feedback and Messaging System

Customers can rate and review workers after service completion. The system includes an internal messaging feature to facilitate communication between customers and workers.

5. External Interface Requirements

5.1 User Interface

The system provides a responsive and modern interface built using React.js and Vanilla CSS. It supports both light and dark modes and is optimized for multiple devices.

5.2 Software Interface

- Backend: Node.js with Express.js
- Database: MongoDB with Mongoose
- API: OpenStreetMap Nominatim API

5.3 Hardware Interface

No special hardware requirements. Any device with internet access can use the system.

6. Functional Requirements

- FR1: The system must allow users to register using a unique email ID.
- FR2: The system must authenticate users using JWT.
- FR3: Workers must not be visible until approved by the admin.
- FR4: Customers must be able to search workers based on skills and location.

- FR5: The system must support booking requests and status tracking.
- FR6: Users must be able to communicate through internal messaging.
- FR7: Customers must be able to submit ratings and reviews.

7. Non-Functional Requirements

7.1 Security

The system ensures security using encrypted passwords, JWT authentication, and role-based access control.

7.2 Performance

The application is optimized for fast loading using Vite and efficient API handling. Response times are maintained under acceptable limits.

7.3 Scalability

The modular architecture allows easy scaling and addition of new features.

7.4 Availability

The system is accessible across multiple devices and browsers with responsive design.

8. Database Design

The system uses a NoSQL database with the following collections:

- User
- Booking
- Feedback
- Message

Each collection is designed to maintain relationships using references.

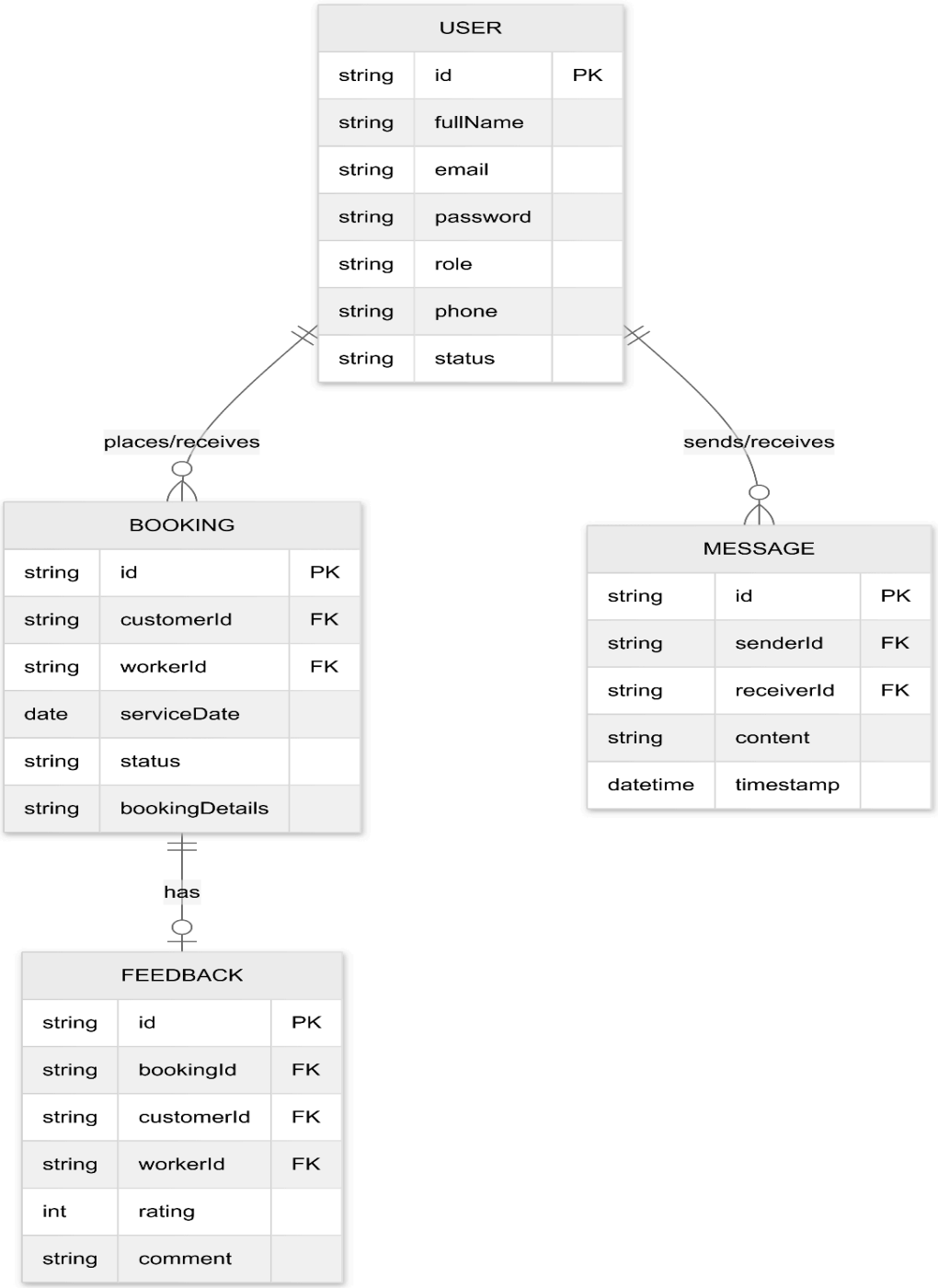


Figure 2: Entity-Relationship Diagram of TrustyHands Database

9. Data Flow Diagrams

9.1 Level 0 (Context Diagram)

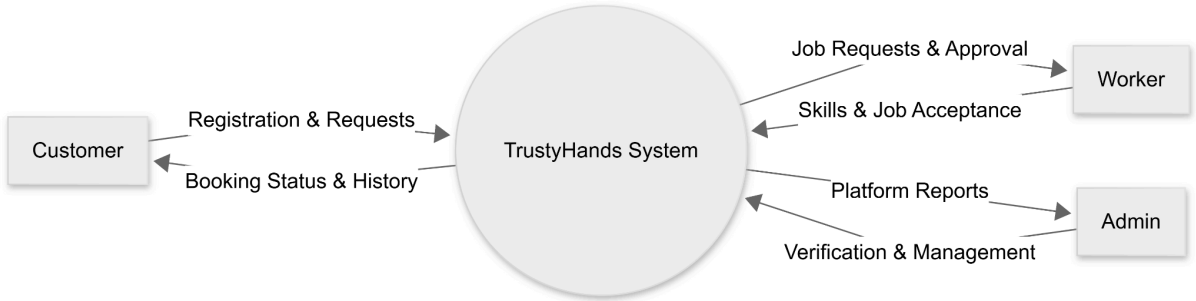


Figure 3: Data Flow Diagram (Level 0 – Context Diagram) of TrustyHands System

9.2 Level 1

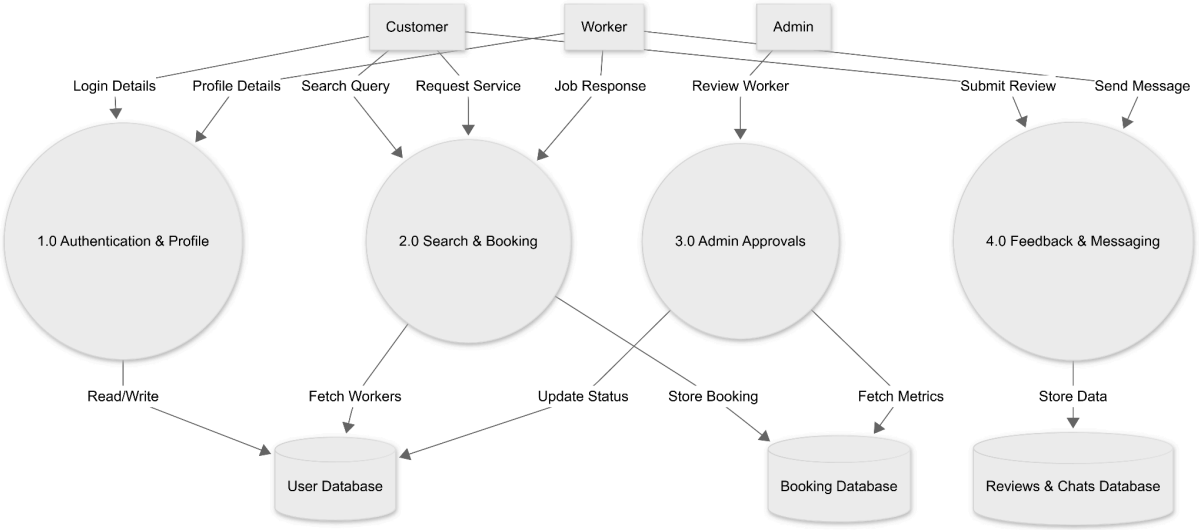


Figure 4: Data Flow Diagram (Level 1) of TrustyHands System

10. Use Case Diagram

This diagram represents interactions between users and the system.

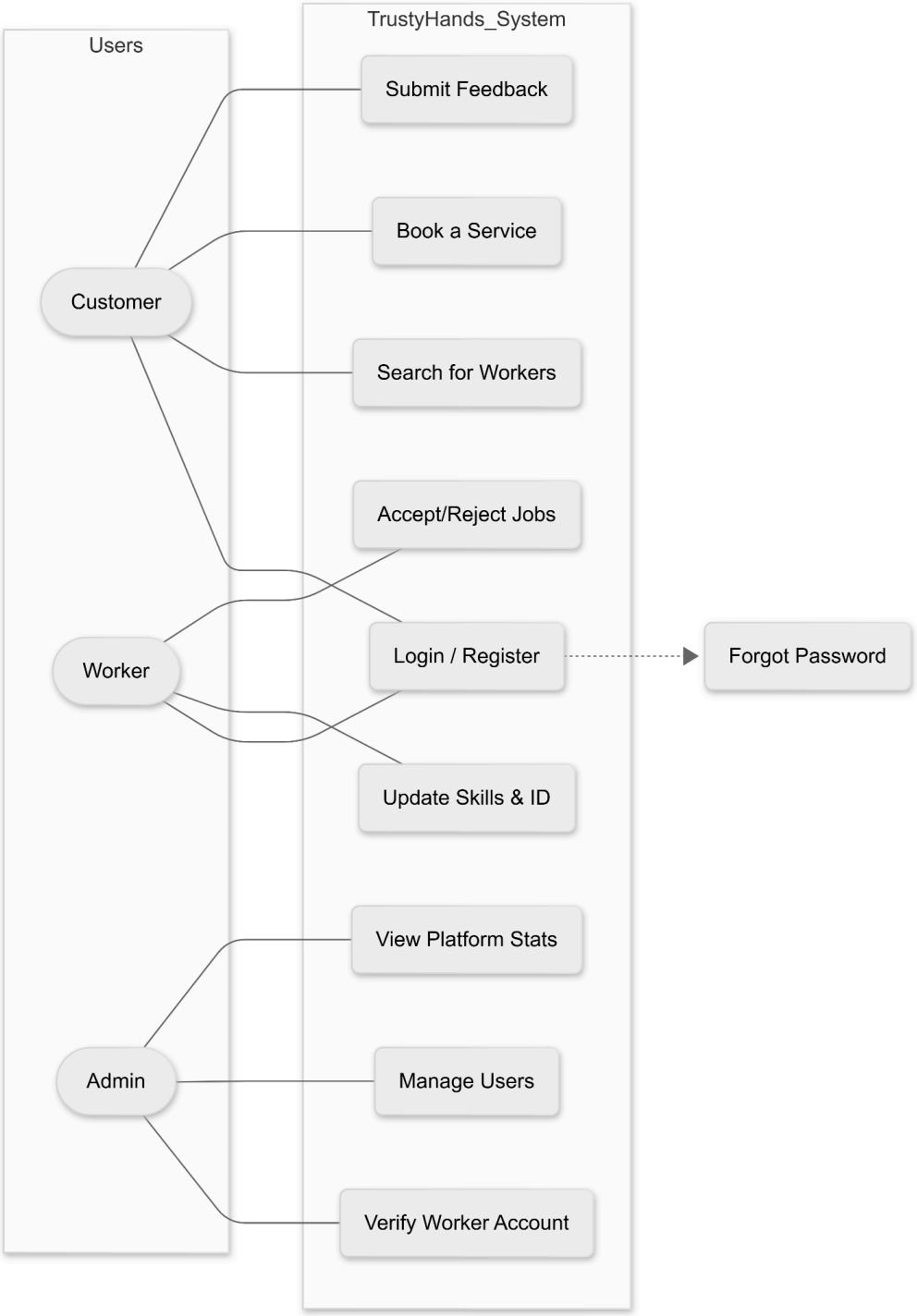


Figure 5: Use Case Diagram for TrustyHands Service Marketplace

11. Sequence Diagram

This diagram illustrates the step-by-step booking process.

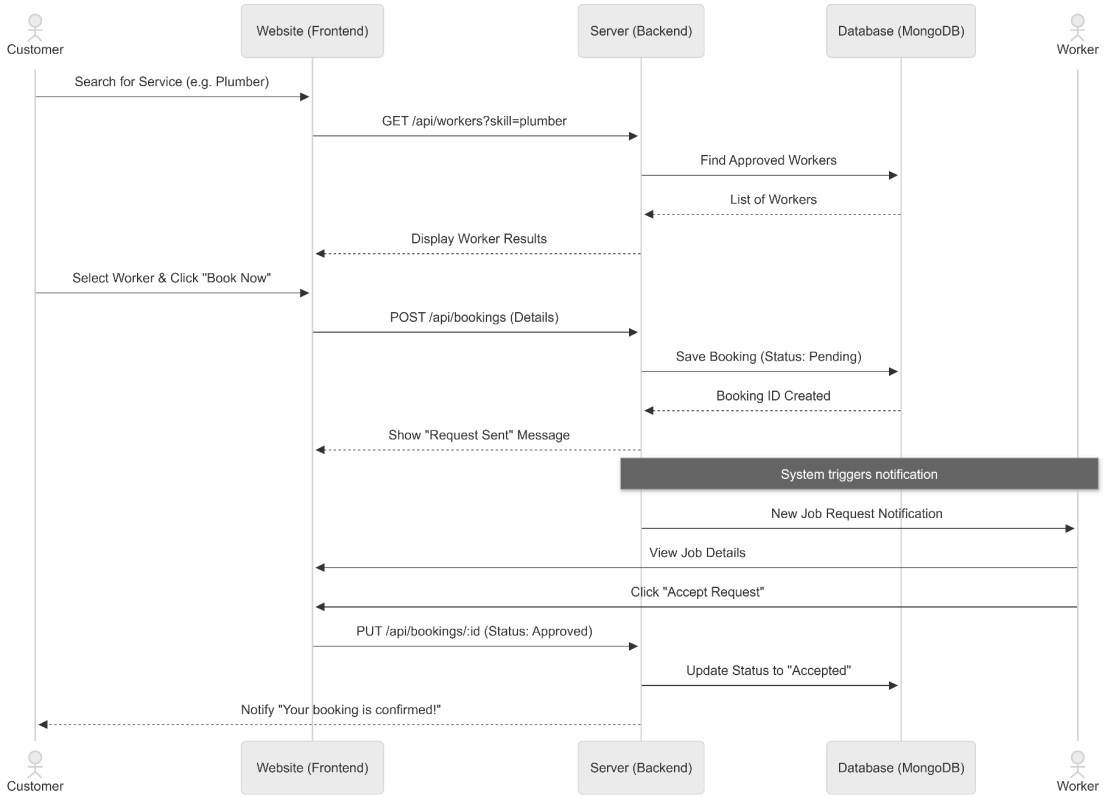


Figure 6: Sequence Diagram for Service Booking Process in TrustyHands

12. Workflow Description

The system follows a structured workflow:

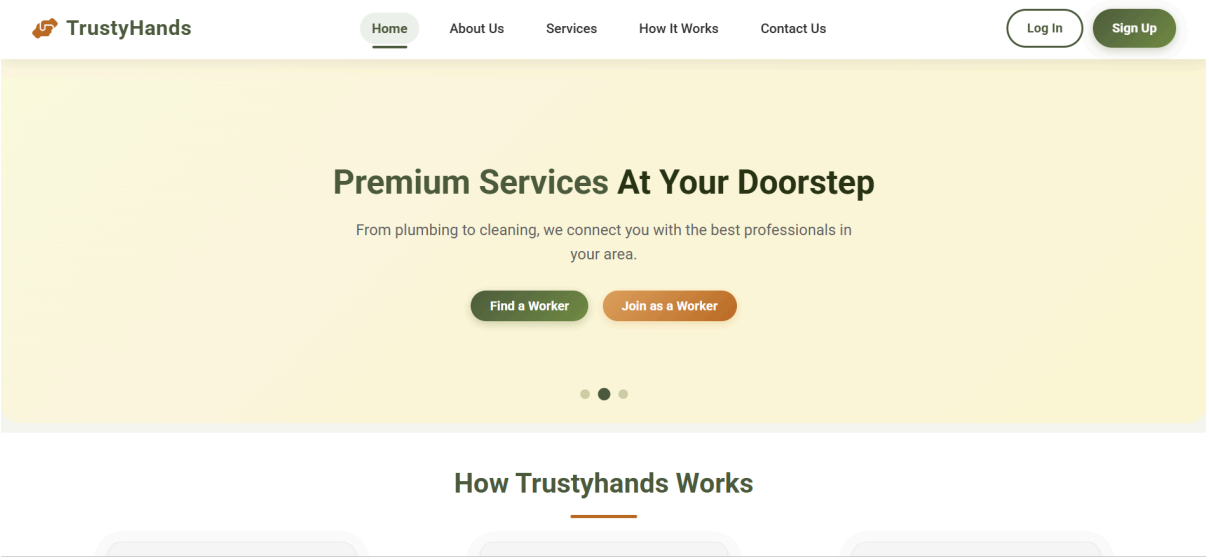
1. User registers as Customer or Worker
2. Worker waits for admin approval
3. Customer searches for services
4. Customer sends booking request
5. Worker accepts or rejects request
6. Service is completed
7. Customer provides feedback

13. Deployment and Maintenance

- Frontend deployed using Vercel/Netlify
- Backend deployed using Render/Heroku
- Database hosted on MongoDB Atlas

The system is designed for easy maintenance with modular code structure and API-based communication.

14 . Results: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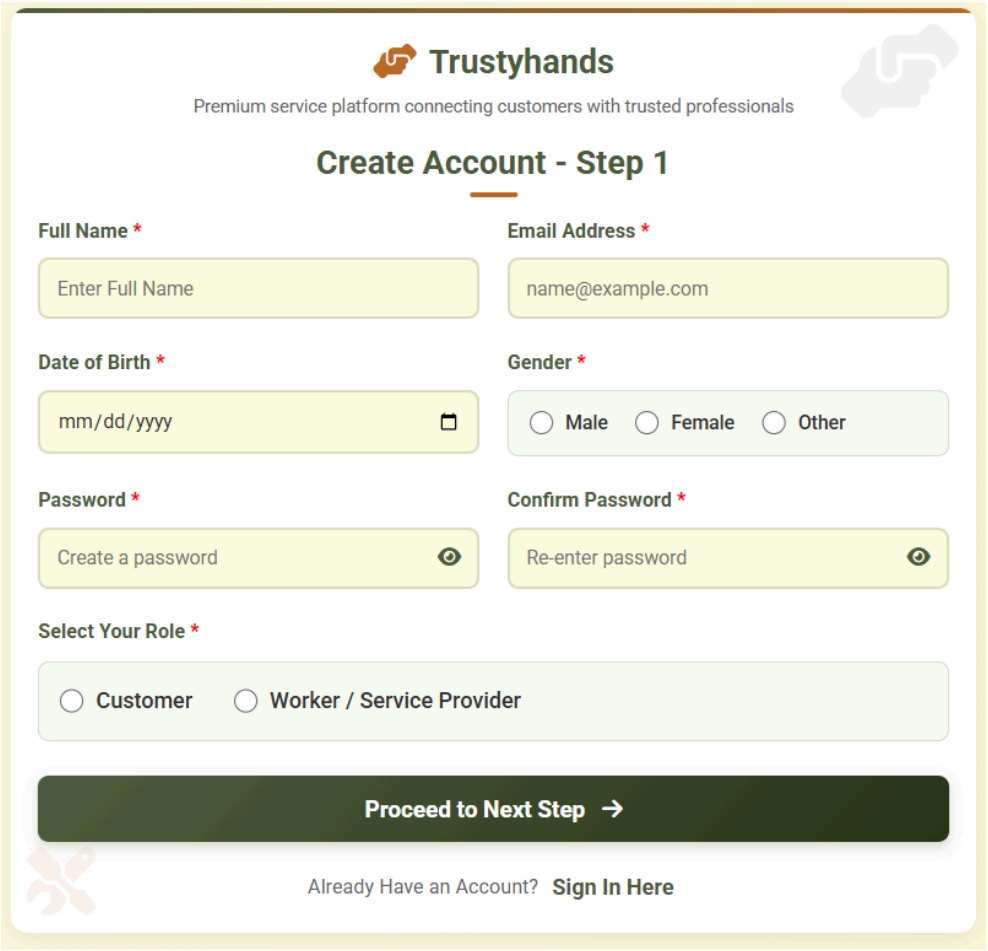
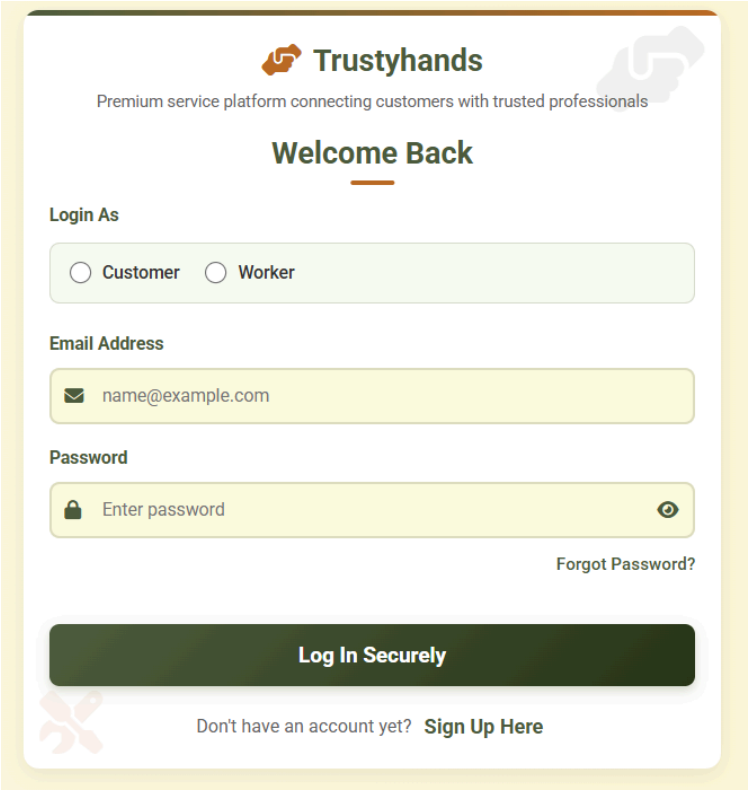
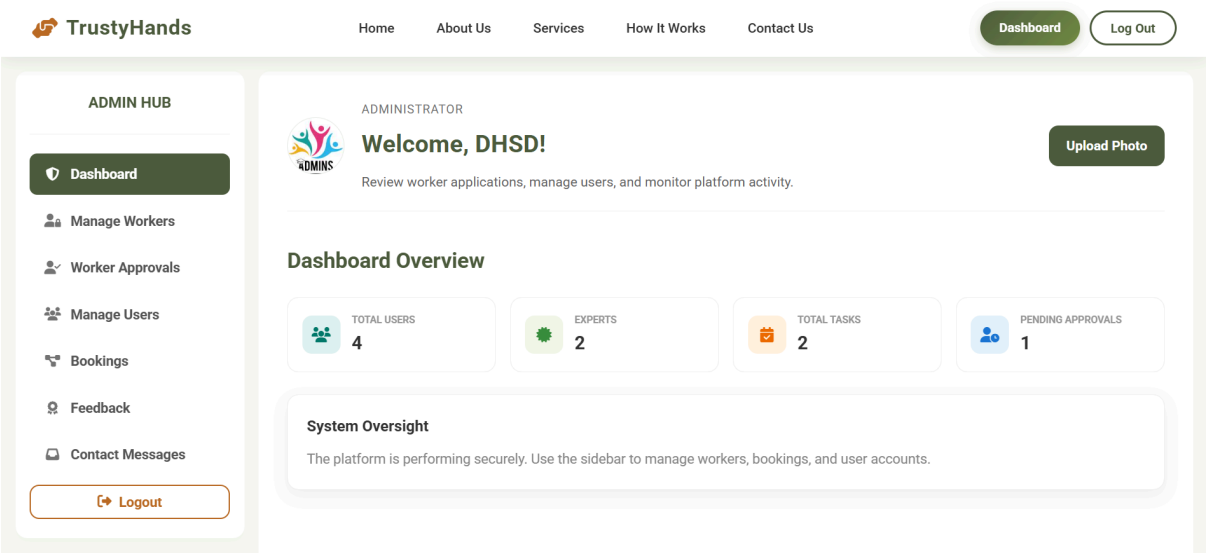
Premium Services At Your Doorstep
From plumbing to cleaning, we connect you with the best professionals in your area.
Find a Worker
Join as a Worker
How Trustyhands Works


Trustyhands
Premium service platform connecting customers with trusted professionals
Create Account - Step 1
Full Name *
Enter Full Name
Email Address *
name@example.com
Date of Birth *
mm/dd/yyyy
Gender *
Male
Female
Other
Password *
Create a password
Confirm Password *
Re-enter password
Select Your Role *
Customer
Worker / Service Provider
Proceed to Next Step →
Already Have an Account? Sign In Here


Trustyhands
Premium service platform connecting customers with trusted professionals
Welcome Back
Login As
Customer
Worker
Email Address
name@example.com
Password
Enter password
Forgot Password?
Log In Securely
Don't have an account yet? Sign Up Here


TrustyHands
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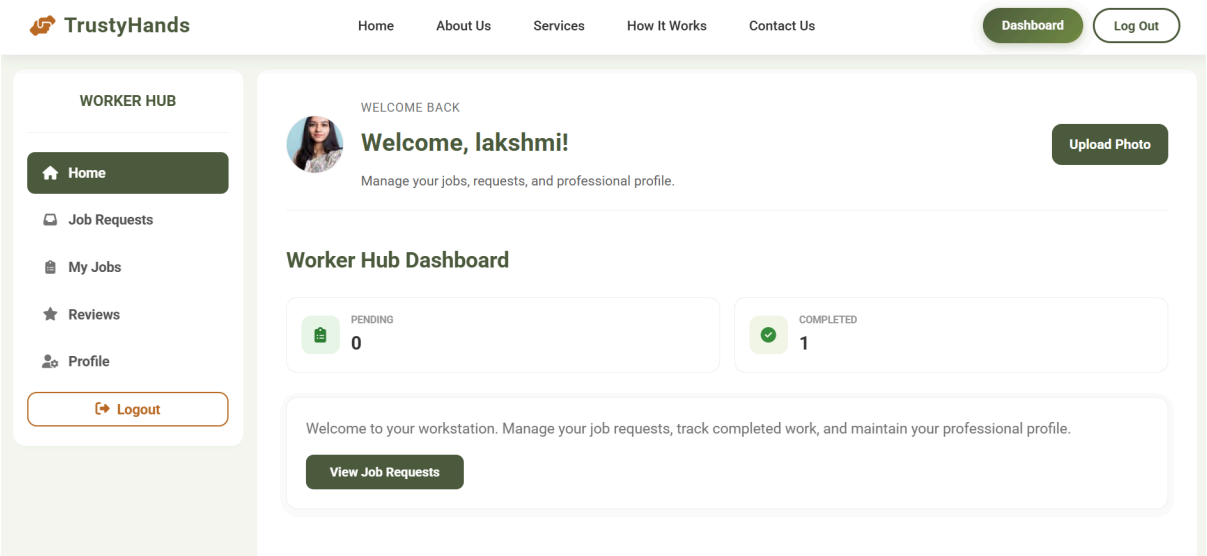
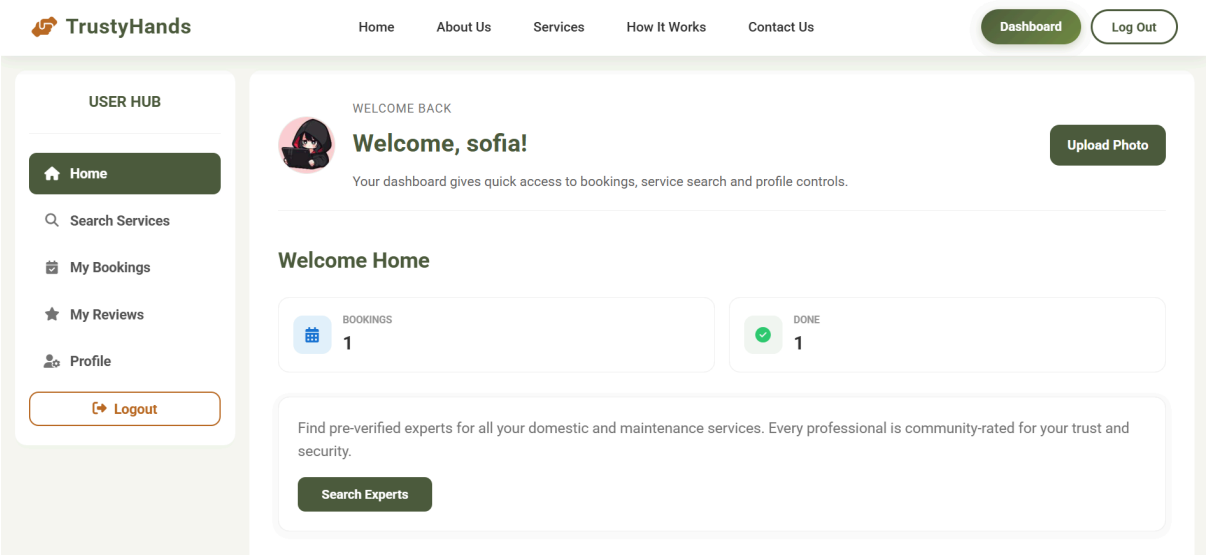
How It Works
Contact Us
Dashboard
Log Out
ADMIN HUB
Dashboard
Manage Workers
Worker Approvals
Manage Users
Bookings
Feedback
Contact Messages
Logout
ADMINISTRATOR
Welcome, DHSD!
Review worker applications, manage users, and monitor platform activity.
Upload Photo
Dashboard Overview
TOTAL USERS
4
EXPERTS
2
TOTAL TASKS
2
PENDING APPROVALS
1
System Oversight
The platform is performing securely. Use the sidebar to manage workers, bookings, and user accounts.





15. Future Scope

The TrustyHands platform can be further enhanced by incorporating advanced features and improvements to increase usability, scalability, and user engagement.

Future enhancements may include:

- Integration of **online payment gateways** such as UPI, credit/debit cards, and wallets for seamless transactions.
- Implementation of **real-time notifications** using WebSockets or Firebase for instant updates on bookings and messages.
- Development of a **mobile application (Android/iOS)** to improve accessibility and user experience.
- Addition of an **AI-based recommendation system** to suggest the best workers based on user preferences and ratings.
- Implementation of **live tracking of service providers** for better transparency.

- Enhancement of security using **multi-factor authentication (MFA)** and advanced fraud detection mechanisms.
- Expansion of services to support **multiple cities and international markets**.

16. Conclusion

The TrustyHands platform successfully addresses the challenge of connecting customers with reliable and verified local service providers through a structured and secure digital system. The application ensures transparency through worker verification, feedback mechanisms, and role-based access control.

The system is designed using modern web technologies and follows a scalable architecture, making it adaptable for future enhancements. With features such as intelligent search, efficient booking workflows, and integrated communication, TrustyHands provides a comprehensive solution for service management.

Overall, the project demonstrates the effective application of software engineering principles, including requirement analysis, system design, and implementation, resulting in a robust and user-friendly platform.